

Questions and Answers

Rubaiyat Alam and Rena Conti

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1. Why do brand drug makers release AGs?

Ans: Segment the market via price discrimination, get a cut of generic's profits/market share.

2. Why do brand drug makers not release AG every time it face generic entry?

Ans: First, the AG would cannibalize the brand drug's sales. Second, the AG entry would lead to generics lowering their prices in response, further increasing the price gap between brand and generics; this may cause brand drug sales to fall even more than without AG entry. Third, it could be due to supply-side reasons such as manufacturing delays, organizational frictions, and legal settlements. Fourth, it could be that many generics enter in the very first period after loss-of-exclusivity, meaning that the AG will make little profit due to intense price competition; this explains why AGs have a strong incentive to enter during the 180-day exclusivity period, as detailed in FTC (2011). In our model we account for all these factors.⁶Section on "Authorized Generics" under Institutional Background.

3. Why do brand drug makers sometimes wait to release AGs?

Ans: Quick response – Manufacturing delays, cost issues, policy/legal uncertainty, inexperience with generic production, organizational/transactional frictions, etc.

More detailed – First, this could be driven by manufacturing delays, organizational frictions, and contracting frictions (if AG production is being outsourced). It may be that the loss-of-exclusivity was unanticipated or there was uncertainty surrounding it, which meant that the brand maker was unprepared for the loss-of-exclusivity when it happens. There could be policy uncertainty such as how the FDA would approve or regulate the new generic entrants, or how a lawsuit might be resolved, that may add further uncertainty to the release decision. It could be that the brand maker is engaging in other generic-detering strategies (such as product-line extensions) which affects its AG timing decision. The brand drug maker may not have any generic subsidiaries, which makes it very difficult to release an AG,

because marketing a nonbrand drug versus a brand drug entail very different skillsets. The marketing of a brand drug focuses on advertising and detailing, negotiating formulary positions by insurer, etc. The marketing of a generic drug focuses on extreme cost efficiency, winning pharmacy auctions (Cuddy (2020)), etc. These are very different skillsets, and for a brand drug, it may not be worth the hassle of getting into the generics business just to release an authorized generic. It could instead outsource its production to another generic firm (for instance, Prasco and Greenstone specialize in bringing authorized generics for other firms into the market) but that involves negotiation cost and profit-sharing.

4. Why don't physicians switch patients over from brand to AG immediately, since they are chemically identical?

Ans: Patients and physicians cannot easily tell whether a generic product is an independent generic or an AG, or whether an AG is available for a molecule-formulation. This is by design, since the AG is meant to compete with generics without cannibalizing the brand drug's sales.

5. Why does the brand maker not always match the price of generics with the brand drug instead of releasing the AG?

Ans: First, there is a price discrimination motive where consumers who like the brand drug are willing to pay a high price for it, so having the expensive brand and inexpensive AG in the market allows the maker to capture both types of consumers. Second, if the generic entrant has to leave the market (e.g. due to legal reasons, patenting issues, technical or manufacturing problems) the brand may not be able to raise its prices back up without facing backlash from buyers. Third, if the brand product's price is lowered, it is hard to justify high prices on line extensions. All the points mentioned above are corroborated in FTC (2011).

6. What causes these "entry delays" for generics?

Ans: Some examples are as follows. First, the most important is the time FDA takes to assess ANDA applications and give approval, which often involves a back-and-forth between the FDA and the maker regarding the production process. Over our dataset's period, the FDA suffered from large backlogs of ANDA applications waiting to be approved, although this has improved. Second, the Hatch-Waxman Act set up a 180-day first-filer-exclusivity provision where the first generic to file for Paragraph IV certification enjoys a 180-day period after entry when other generics cannot enter; thus, a rival becoming the first-filer stops other generics from entering the market for 180 days. Third, final approval by the FDA requires inspections to meet CGMP (Current Good Manufacturing Practice Regulations) at scale, and they can fail. Furthermore, for certain types of molecules and formulations, the generic

maker can face manufacturing problems when setting up production lines for FDA approval, delaying approval further. Fourth, unrelated inspections for the first-filer can fail, which then cascades to interrupting final approvals for all subsequent filers. Fifth, there have also been reports of brand drug makers delaying providing samples for generic makers to copy in an effort to slow down ANDA approvals. Finally, there may also be information issues due to legal or technical reasons such that the generic maker cannot correctly anticipate the date of LOE, leading to delays in filing and hence approval.

7. What about rebates that drug makers pay to insurers?

Ans: Generic makers do not pay rebates, but brand drugs may, creating potential measurement errors for brand drug prices. To address this, as a robustness check we rerun our demand estimation allowing brand drugs to provide 30% rebates (following the findings in Kakani et al. (2020) and Kakani et al. (2022)), and obtain very similar results.

8. Why define market at the narrow molecule-formulation level, rather than say at the therapeutic class level?

Ans: First, we include a molecule-formulation fixed effect in the demand function to account for other molecule-formulations in the same therapeutic class. Second, similar market definitions have been used in various antitrust proceedings and empirical papers. Third, this allows us to capture substitution patterns between bioequivalent products more accurately. Fourth, the brand drug's patent and generic approval holds at the molecule-formulation level. Fifth, one drug can be in multiple therapeutic classes, making modeling therapeutic classes as markets unclear. Finally, evidence suggests that while there can be substitution across molecule-formulations within the same therapeutic class, the substitution is usually from a molecule-formulation without a generic to a molecule-formulation with one.

9. How come you have only 241 molforms?

While our dataset has comprehensive coverage across all prescription drugs sold in the US, we look at a smaller subset of drugs for our project. We select molecule-formulations that saw its first generic entrant after 2004 and before 2016, which are not biologics, for which we can identify presence or absence of AG with certainty, and those that are not surrounded by strong external circumstances (e.g. media/public outrage, serious or repeated lawsuits). This leaves us with 241 molecule-formulations.

10. How often are AGs released even when there are no generics?

Ans: Almost never. As corroborated by the FTC (2011), AGs are almost always released in response to generic entry. In our data, we see one such case.

11. What about brand drug being pulled from the market as an additional strategic choice?

Ans: Of our 241 molecule-formulations, 31 see the brand drug exiting. For these 31 markets, three see an AG released. Given how infrequently the brand drug exits the market, and the computational burden it will add to our estimation, we do not model brand drug exit decisions.

12. Regarding the variation in generic entry timing, how about these alternative story – all generics get approved immediately after LOE, and stagger launch over time due to manufacturing frictions or strategic timing decisions?

Ans: Wang et al 2022 uses exact dates from the Orange Book on when a generic receives FDA approval, and finds patterns very similar to ours. So generics are launching very soon after receiving FDA approval. This logically makes sense, as they want to start recouping their sunk cost of entry, and the earlier periods of the market are the most profitable.

13. Why have brand-specific time trend in the demand model?

Ans: The time trend captures the following issues. Brand drug makers can sometimes take actions that slow down generic adoption. This involves actions such as product hopping, getting late-stage-lifecycle approvals for new indications, etc. These actions hinder consumers and pharmacists from switching to generic drugs and getting around automatic substitution laws. While our model does not explicitly account for such actions, our demand specification contains a brand-specific time trend that captures the idea that the brand drug loses its appeal over time as the maker runs out of such tactics. As expected, we find that the brand drug becomes less attractive over time, possibly because such tactics become less effective over time.

14. Why have this "two-stage" demand model?

This two-stage model has two advantages. First, it reflects institutional reality where the patient/provider/insurer decides which molecule-formulation to purchase and whether to buy the brand product, then the pharmacist or procurement officer decides which specific product to supply to the patient. (For instance, pharmacies choose which generics to stock, as described in Cuddy (2020).) Second, for estimation and counterfactuals, this matches our data patterns better than single-stage discrete-choice specifications.

While our demand model is somewhat different from other discrete-choice settings, note that our project aims to study entry decisions; as long as we correctly predict prices and per-period profits, our central arguments work.

15. What about no-AG settlements? What about product-line extensions and other strategic actions by the brand drug maker?

Ans: Since no-AG settlements unobserved, these are subsumed in the AG choice-specific logit shocks. So are product-line extensions, although that's a possible future paper to look at that can use the current paper's methodology.

16. How do generics filing paragraph IV certifications or presence of pay-for-delays feature in the model?

The decisions to file Paragraph IV certification and negotiate any settlements (such as pay-for-delays or no-AGs) are exogenous to our model. More specifically, we think of Paragraph-IV certifications and pay-for-delays determining when a molecule-formulation is open for generic entry. Our model takes this opening date as exogenously given. Regarding first-filer exclusivity, our model assumes each generic assigns some probability that they are the first-filer, or that they have to wait 180 days to enter because a rival achieved the first-filer status instead. These probabilities are reflected in our calculation of generic entry rates.

In general, we do not endogenize decisions surrounding patent lawsuits and settlements, and instead allow drug makers to anticipate them with some probability. Our paper focuses on the periods of and following generic entry. These are simplifications driven by data limitations and computational concerns, and they allow us to tractably model the generic entry decision.

Note that FTC (2011) finds most AGs are released outside the 180-day exclusivity, and often for drugs that did not see any patent challenges; see Appendix J.3. Note also that FTC (2011) argues AGs are unlikely to deter Paragraph IV certifications.

17. How are the FDA approval rates/generic entry rates estimated?

Ans: Directly from the generic entry rates observed in the data.

18. How is market size calculated?

Ans: For market size calculation of a molecule-formulation, we find the maximum total units sold in any quarter for that molecule-formulation over our dataset, then scale it up by a constant. This constant is the same for all molecule-formulations. In our baseline estimation, the constant is set to be 4. In the robustness checks, we set it to be 2, 3, and 5, and get very similar results for the demand estimation.

19. Don't patients make repeat purchases of the same drug? How does that feature here?

(or) Don't physicians and PBMs and pharmacies assist in drug choice? How does that feature here? Ans: Not really the right way to think of our demand model. Our demand model says:

for some average price of drug in US, what is the market share it expects to get. This is the overall demand curve facing the firm. You can think the firm uses this to make entry decisions, and after doing so begins the finer negotiation across formularies and insurers. All the pharmacy/PBM/repeat purchase stuff is subsumed in this overall demand model, rather than being modeled explicitly.

20. Don't generics anticipate LOE and file for ANDA earlier?

Ans: Some do, and while we do not model that decision to "file early" specifically, it is baked into the estimation of the approval probabilities. In effect, when generics apply, it is more likely that they get approved earlier rather than later in the market. This brings in the idea that some generics may be applying early so that they get approval earlier.

21. How does generic collusion factor into this?

Ans: Demand estimation remains unaffected.

Supply estimation - This alleged conduct occurred from late 2013 onwards; our dataset's coverage is from 2004-2016. As a result, we rerun the price prediction regression by dropping years 2013 onwards, and obtain very similar results.

22. You're assuming ex-ante identical generic entrants, how plausible is that?

Ans: Simplifying assumption that significantly eases computation, particularly from multiple equilibria.

23. Who exactly is paying the "price" in your model?

A: Price calculated as total revenue received by drug manufacturer divided by total units sold. Can think of this as the price charged by the drug manufacturer to the entire healthcare system.